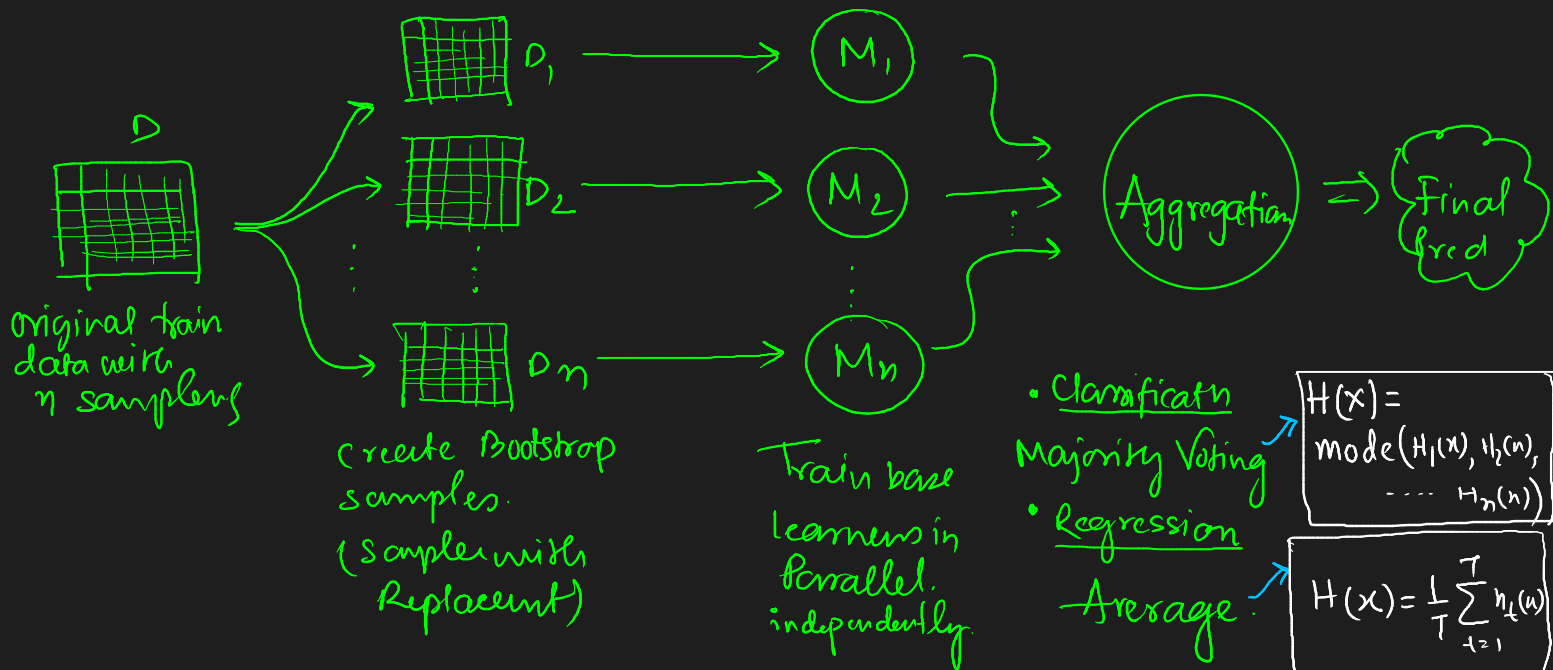
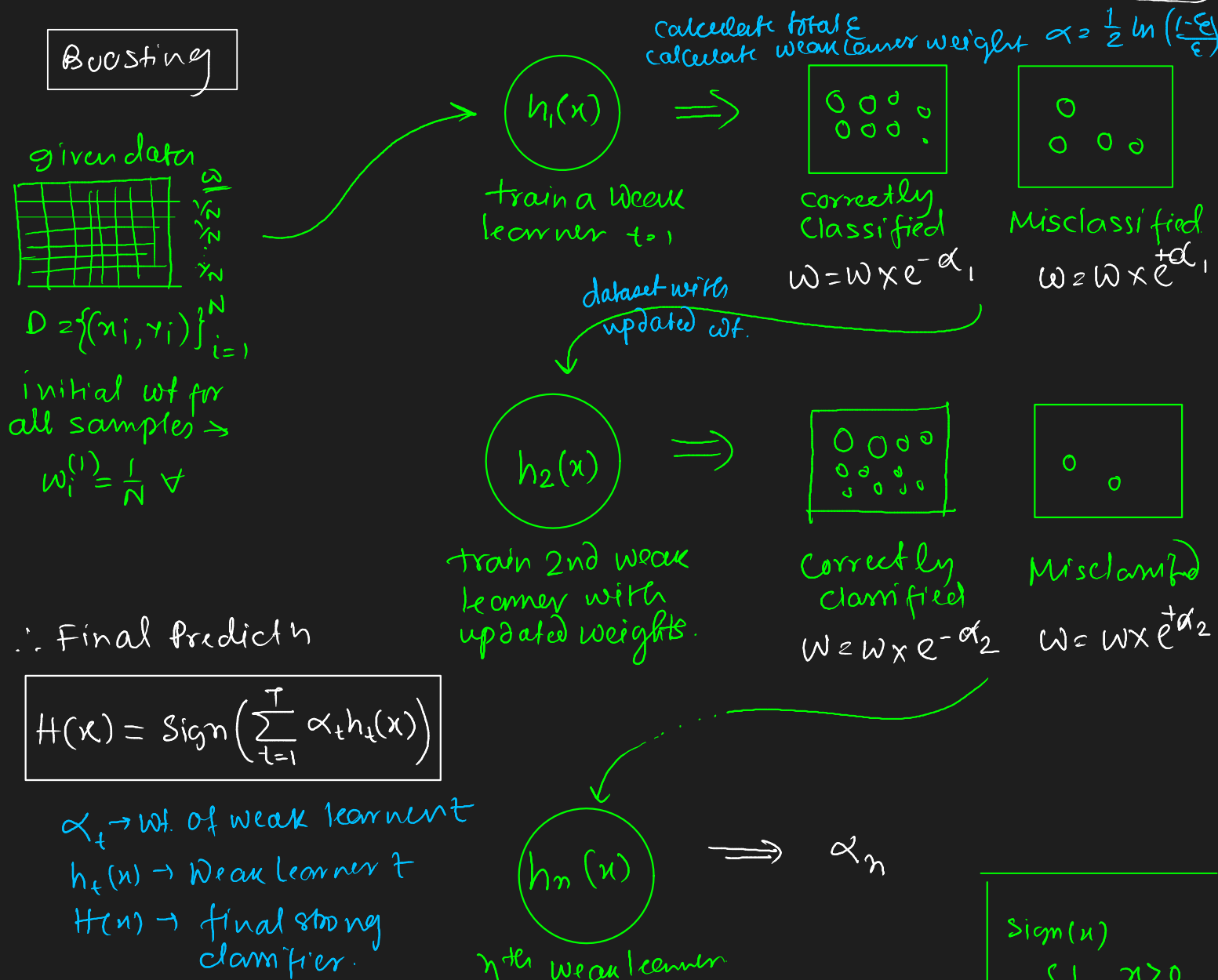


# ML Revision.

## Bagging



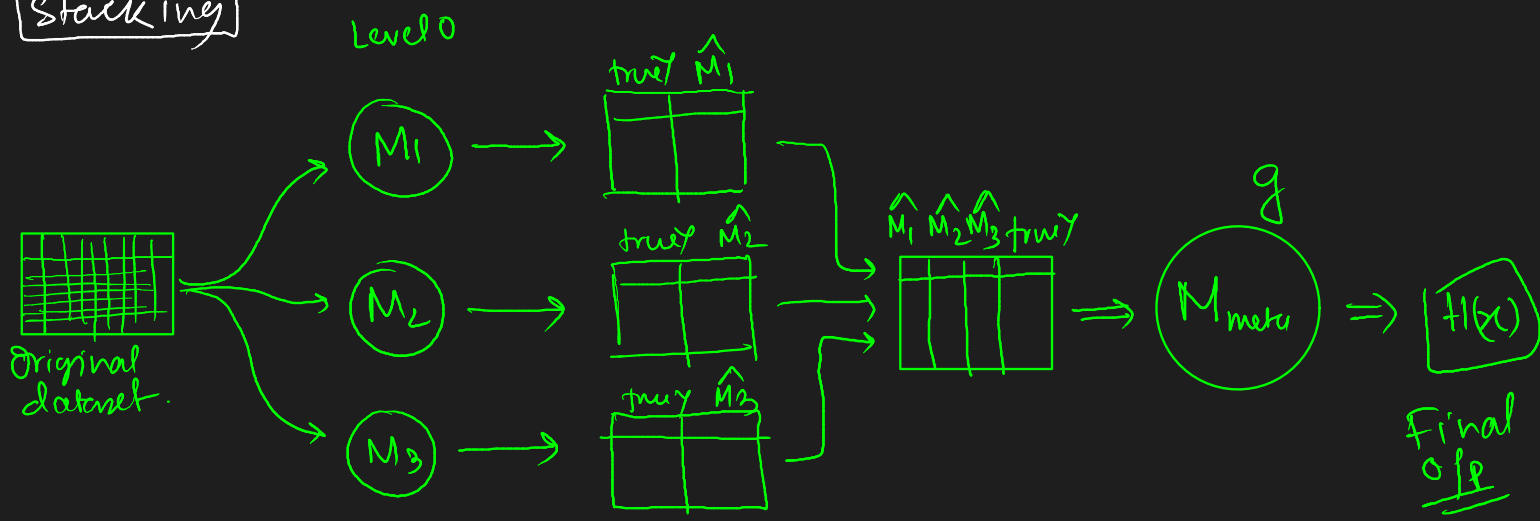
## Boosting



Sign(x)

$$= \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

# Stacking



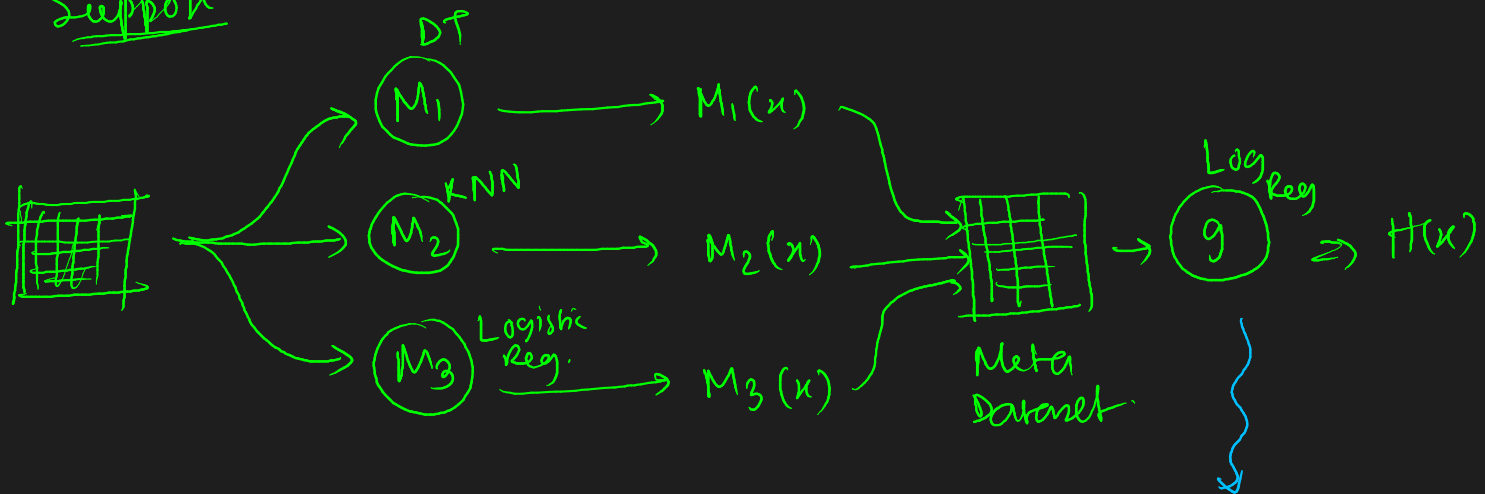
Final Pred<sup>h</sup>

$$H(x) = g(h_1(x), h_2(x), h_3(x))$$

$g(\cdot)$   
= Meta Learner

Pred<sup>h</sup> of  $M_1$     Pred<sup>h</sup> of  $M_2$     Pred<sup>h</sup> of  $M_3$

## Support



Suppose  $g$  learned  $w_1, w_2, w_3$

$$H(x) = w_1 M_1(x) + w_2 M_2(x) + w_3 M_3(x) + b$$

# PCA workflow

Standardize data

$$Z = \frac{x - \mu}{\sigma}$$

Compute cov. matrix

$$\text{Cov}(X, Y) = \frac{\sum (\bar{x} - \bar{x})(\bar{y} - \bar{y})}{n-1}$$

$$C = \begin{bmatrix} \text{Cov}(X, X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Cov}(Y, Y) \end{bmatrix}$$

find Eigen val. & Eigen vector

$$A v = \lambda v$$

Eigen val. (Max var. value)  
Eig vec. (dirn of Max var.)

Transform Data

$$Z = W X$$

Transform data

Eigenvector Matrix

Original data

Select PC

$$\lambda_1 > \lambda_2 > \lambda_3$$

$pc_1, pc_2$

$$\text{Var}(X) = \frac{\sum (x - \bar{x})^2}{n}$$